
Fine-Tune Once, Reuse Across Models: Bayesian Task-Update Factors and Approximations

Siyang Guo¹ Junbo Wang^{1 2} Zibin Zheng³

Abstract

As pre-trained models evolve rapidly, transferring fine-tuning knowledge to updated models without retraining has become a critical challenge. Most existing methods reuse parameter updates, yet the same dataset can induce substantially different updates across base models due to mismatched local loss landscapes, making such transfer unstable. We instead adopt a Bayesian-updating perspective: a base model defines a prior, while fine-tuning contributes a task-update factor that is prior-agnostic, thereby making it feasible to reuse the update across base models. Specifically, we formalize a reusable task-update factor by requiring *invariance across base models* and *a fixed-dimensional parameterization*. Our main theoretical result shows that such reusable factors exist when the variational family is a half-space, and it is already maximal among convex families. In particular, an ideal regime arises when the priors and their Bayesian posteriors remain within a shared exponential family, as it always admits a reusable update factor. Building on this existence, we propose **BTransfer**¹, which extracts a reusable task-update factor from a single fine-tuning run and applies it to a new prior. For deep networks, we implement **BTransfer** with a “lift–transfer–return” pipeline: 1) lift model parameters to distributions; 2) transfer the extracted task-update factor in the exponential family distributions; and 3) return the updated posterior distribution to parameter space. Extensive experiments demonstrate that our approach effectively

reuses fine-tuning knowledge across models without post-training.

1. Introduction

The emergence of Large Language Models (LLMs) has greatly accelerated the development of general-purpose intelligent systems, gradually becoming a unified backbone for a wide range of contemporary AI applications (Radford et al., 2021; Team et al., 2023; Zheng et al., 2023; Tong et al., 2024a; Naveed et al., 2025). By performing lightweight fine-tuning on a pre-trained LLM using downstream task data, one can rapidly adapt it to a wide range of specific applications without training from scratch (Houlsby et al., 2019; Hu et al., 2022; Gao et al., 2024; Liu et al., 2024b; Weyssow et al., 2025). Pre-trained models fine-tuned on task-specific datasets already have demonstrated outstanding performance across diverse scenarios, including natural language processing, computer vision, and multi-modal learning (Ouyang et al., 2022; Chen et al., 2023; Huang et al., 2023; Chung et al., 2024).

Despite the strong empirical performance of fine-tuning, the continual evolution of pre-trained models introduces new challenges (Touvron et al., 2023a;b; Grattafiori et al., 2024). In particular, when the underlying LLM is replaced by a more powerful pre-trained model, the task-specific knowledge encoded in the fine-tuned checkpoints of the old model is often difficult to directly reuse. Fine-tuning the new model on the same task can alleviate this issue, but typically incurs substantial computational cost (Kim et al., 2023). Moreover, the original data for fine-tuning may be inaccessible due to privacy, legal, or storage constraints (Zhu et al., 2021; Liu et al., 2024a). As a result, reliably transferring existing fine-tuning knowledge from an old pre-trained model to a new one without additional training has become a key challenge.

Existing works represent task-specific knowledge as a parameter update, i.e., the parameter difference between the fine-tuned model and the pre-trained model (also called a task vector) (Cao et al., 2025; Gu et al., 2025; Rinaldi et al., 2025b). They reuse this knowledge by adding the resulting update to the parameters of a new pre-trained model. Later

¹School of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China ²Provincial Key Laboratory of Fire Science and Intelligent Emergency Technology, Guangdong, China ³School of Software Engineering, Sun Yat-sen University, Zhuhai, China. Correspondence to: Junbo Wang <wangjb33@mail.sysu.edu.cn>.

Proceedings of the 43rd International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026. Copyright 2026 by the author(s).

¹Code is available at [BTransfer](#).

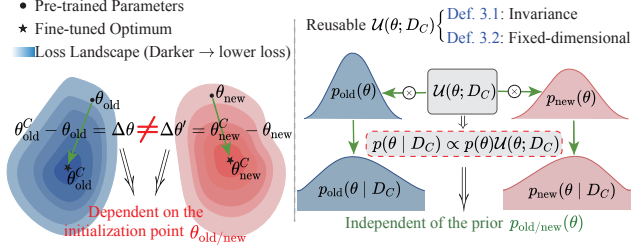


Figure 1. Left. Fine-tuning on the same dataset D_C from different pre-trained checkpoints yields misaligned task updates due to differing local loss landscapes. **Right.** Interpreting fine-tuning as a Bayesian update $p(\theta | D_C) \propto p(\theta)U(\theta; D_C)$, the task-update factor depends only on D_C (not the prior), and can therefore be reused across priors to obtain their posteriors.

variants incorporate a re-basin step: leveraging permutation symmetries in neural networks, they search within the equivalence class of permuted update vectors to find the one that best matches the new model’s parameter indexing, thereby reducing coordinate mismatch (Rinaldi et al., 2025a). Overall, these methods implicitly rely on a strong premise: the same dataset induces similar parameter updates in both the old and the new pre-trained models (Ilharco et al., 2023; Zhang et al., 2024).

However, this premise is brittle in practice. Different pre-trained models can exhibit substantially different local loss landscapes—including gradient directions, curvature, and nearby optima—so the same dataset can induce updates that are not well aligned across models. When these landscapes differ significantly, applying an update extracted from previous fine-tuning to a new model may fail to reflect the true update from direct fine-tuning, leading to unstable transfers.

This motivates our central question:

Can we construct a task update that is invariant across pre-trained models?

Our key insight is to shift from parameter updates to a Bayesian-updating perspective: a base model specifies a prior, while fine-tuning contributes a task-update factor that is prior-agnostic. If this factor admits a compact representation, we can compute it once from a single fine-tuning run and reuse it across base models (see Fig. 1).

Mathematically, we formalize a **reusable task-update factor** by requiring: i) invariance across base models, i.e., it is determined solely by the dataset and does not depend on the particular pre-trained model; and ii) a fixed-dimensional parameterization whose dimension does not grow with the number of training examples, enabling constant storage and transfer cost. Such a factor is not guaranteed to exist for

general Bayesian updates, so we model fine-tuning via variational inference and restrict the solution to a structured variational family. We then show that a reusable task-update factor satisfying both properties exists when the variational family is chosen to be a convex half-space, and this is already maximal among admissible convex families. This yields a structural characterization of the variational families that can support exact, reusable task updates. In particular, an ideal regime arises when the priors and their Bayesian posteriors remain within a shared exponential family, which guarantees the existence of a reusable update factor.

Building on this characterization, we propose **Bayesian Task Update Transfer** (BTransfer). We first develop BTransfer in this idealized exponential-family setting, where it explicitly recovers a reusable task-update factor from a single fine-tuning run and applies it to a new prior to obtain the corresponding Bayesian posterior. For deep networks, BTransfer is implemented via a three-step “lift–transfer–return” procedure: 1) lift model parameters to distributions using a Laplace approximation; 2) transfer the recovered task-update factor in the exponential-family distributions; and 3) return the updated posterior distribution to parameter space. Overall, BTransfer transfers fine-tuning knowledge without re-training by computing a reusable update factor once from a single fine-tuning run and reusing it across future pre-trained models.

In summary, our contributions are as follows:

1. We formalize the concept of a reusable task-update factor and establish a structural characterization of when it exists: we show that such a factor exists when the variational family is a half-space, and this is already maximal admissible among convex families (Sec. 3.1).
2. We propose BTransfer, which extracts a task-update factor from a single fine-tuning run on an old base model and applies it to the prior induced by a new base model, yielding the corresponding updated posterior (Sec. 3.2).
3. We instantiate BTransfer for deep networks via a “lift–transfer–return” procedure, and extensive experiments on language understanding and generation tasks show that it effectively transfers fine-tuning knowledge without retraining (Secs. 3.3 and 4).

2. Preliminary

Variational Perspective of Fine-Tuning. Given a fine-tuning dataset $D_C := \{(x_i, y_i)\}_{i=1}^N$ and a base model parameterized by $\theta \in \mathbb{R}^m$, we model fine-tuning as a variational inference problem. Specifically, we approximate the posterior $p(\theta | D_C)$ by a tractable distribution $q \in \mathcal{Q}$ by

minimizing

$$\min_{q \in \mathcal{Q}} \mathbb{E}_{q(\theta)} [-\log p(D_C | \theta)] + \text{KL}(q(\theta) \| p(\theta)). \quad (1)$$

Here, $p(\theta)$ denotes the prior specified by a pre-trained model. The optimal solution (called **variational posterior**) to Eq. (1) admits the form

$$q^*(\theta) = \frac{\mathcal{U}(\theta; D_C) p(\theta)}{Z(D_C; p)},$$

where $\mathcal{U}(\theta; D_C)$ is a task-update factor that encodes the dataset-induced update, and $Z(D_C) := \int \mathcal{U}(\theta; D_C) p(\theta) d\theta$ is the normalizing constant that ensures q^* integrates to one.

Fine-tuning Transfer. Within the above framework, we study fine-tuning transfer across base models. Consider two priors induced by two pre-trained checkpoints, $p_{\text{old}}(\theta)$ and $p_{\text{new}}(\theta)$. Starting from $p_{\text{old}}(\theta)$, we fine-tune on the task dataset D_C by solving Eq. (1), obtaining the variational posterior:

$$q_{\text{old}}^*(\theta) \propto \mathcal{U}(\theta; D_C) p_{\text{old}}(\theta).$$

The goal of fine-tuning transfer is to transfer this fine-tuning knowledge to the new prior without additional training. Specifically, we apply the same update factor to the new prior to form

$$q_{\text{new}}(\theta) \propto \mathcal{U}(\theta; D_C) p_{\text{new}}(\theta),$$

and seek $q_{\text{new}}(\theta)$ to match the variational posterior $q_{\text{new}}^*(\theta)$ from directly fine-tuning the new model on D_C .

If $\mathcal{Q}$ is the unconstrained family of all probability distributions Δ , then $q_{\text{old}}^*(\theta)$ coincides with the Bayesian posterior $p(\theta | D_C)$. In this case, the update factor reduces to the likelihood: $\mathcal{U}(\theta; D_C) = p(D_C | \theta) \propto \prod_{(x_i, y_i) \in D_C} p(y_i | x_i, \theta)$, which is prior-agnostic. Thus, applying the same factor to the new prior $p_{\text{new}}(\theta)$ yields the optimal solution, i.e., $q_{\text{new}}(\theta)$ coincides with $q_{\text{new}}^*(\theta)$ up to normalization. Unfortunately, representing the likelihood typically requires access to the entire dataset D_C , so its storage and communication cost grows with $|D_C|$, making it impractical as a transferable update in our setting. In the next section, we formalize what it means for a task-update factor to be **reusable**, and identify conditions under which such a factor can exist and how to reuse it across pre-trained models.

3. Methodology

To achieve fine-tuned knowledge transfer, our core idea is to identify a reusable task-update factor: this factor is extracted from a single fine-tuning run on the old base model and applied to the prior induced by the new base model to obtain its updated posterior without re-training. In the following sections, we i) formalize the reusable task-update factor

and derive conditions for its existence (Sec. 3.1), ii) present BTransfer (Sec. 3.2), and iii) detail its instantiation for deep networks (Sec. 3.3).

3.1. Reusable Task-Update Factor

Recall that fine-tuning on D_C w.r.t. a prior $p_i(\theta)$ via Eq. (1) induces a variational posterior of the form $q_i^*(\theta) \propto \mathcal{U}(\theta; D_C) p_i(\theta)$. Let $\mathcal{P} = \{p_i\}_{i=1}^N$ denote a collection of priors of interest, where each prior corresponds to a pre-trained model. We argue that a reusable task-update factor should satisfy the following two properties.

Definition 3.1 (Invariance across base models). Given a dataset D_C , we say that a task-update factor $\mathcal{U}(\theta; D_C)$ is invariant across $\mathcal{P}$ if for every prior $p_i \in \mathcal{P}$, the optimal solution to Eq. (1) can be written as

$$q_i^*(\theta) \propto \mathcal{U}(\theta; D_C) p_i(\theta).$$

The invariant property across models ensures that the influence of D_C on the priors can always be expressed through the same update factor $\mathcal{U}(\theta; D_C)$.

Definition 3.2 (Fixed-dimensional Parameterization). Given a dataset D_C , a task-update factor $\mathcal{U}(\theta; D_C)$ admits a fixed-dimensional parameterization if there exists a finite-dimensional function space $\Phi_d := \text{span}\{\phi_1(\theta), \dots, \phi_d(\theta)\}$ independent of D_C such that²

$$\log \mathcal{U}(\theta; D_C) \in \Phi_d.$$

If $\mathcal{U}(\theta; D_C)$ satisfies Definition 3.2, then there exists a parameter vector $\lambda(D_C) \in \mathbb{R}^d$ and basis functions $\phi(\theta) = (\phi_1(\theta), \dots, \phi_d(\theta))$ such that $\mathcal{U}(\theta; D_C) = \exp(\lambda^\top \phi(\theta))$. That is, the influence of D_C is fully encoded by a d -dimensional vector λ , whose dimension remains fixed and does not grow with the dataset size.

Although the two properties above appear natural, we will show that a reusable task-update factor—i.e., one that is both *invariant across base models* (Definition 3.1) and admits a *fixed-dimensional parameterization* (Definition 3.2)—does not exist in general without restricting the variational family. Before stating our main result, we introduce several notations. Fix basis functions $\phi(\theta)$ and a parameter vector λ , and define the resulting distribution for prior p_i as

$$\hat{q}_i(\theta) := \frac{\exp(\lambda^\top \phi(\theta)) p_i(\theta)}{Z_i},$$

where Z_i is the normalizing constant. We also define the auxiliary function

$$g(\theta) := \lambda^\top \phi(\theta) - \log p(D_C | \theta),$$

²Parameterize $\log \mathcal{U}$ to ensure $\mathcal{U}(\theta; D_C) > 0$.

which measures the discrepancy between a finite-dimensional update and the log-likelihood term.

The following theorem provides a sufficient structural condition for the existence of a reusable task-update factor.

Theorem 3.1. *Let Δ denote the set of all probability distributions over θ for which $\mathbb{E}_q[|g|] < \infty$. Assume there exist (ϕ, λ) and a constant c such that $\mathbb{E}_{\hat{q}_i}[g(\theta)] = c$ for all $p_i \in \mathcal{P}$. If we define the variational family in Eq. (1) as $\mathcal{Q}_c := \{q \in \Delta : \mathbb{E}_q[g(\theta)] \geq c\}$, then:*

1. **Existence.** *There exists a reusable task-update factor, namely $\mathcal{U}(\theta; D_C) = \exp(\lambda^\top \phi(\theta))$.*
2. **Maximality.** *For any strictly larger convex set $\mathcal{Q}' \supsetneq \mathcal{Q}_c$, this reusable task-update factor no longer exists.*

The theorem implies that selecting the variational family $\mathcal{Q}_c$ guarantees the existence of a reusable task-update factor satisfying both properties. However, this existence comes at a price: within the class of convex sets, $\mathcal{Q}_c$ is already the maximal admissible variational family. Intuitively, a single finite-dimensional update factor can uniformly capture the dataset’s effect on different priors only when the variational posterior is constrained to a sufficiently controlled family. Once this constraint is relaxed, the variational problem may attain more expressive—potentially smaller objective value—solutions that depend on the particular prior $p_i \in \mathcal{P}$. As a result, the optimal posteriors may drift across priors in a complex way that cannot be described by this fixed-dimensional update factor.

Illustrations of Thm. 3.1. We present two corollaries that clarify how the maximal variational family $\mathcal{Q}_c$ changes. The first relates $\mathcal{Q}_c$ to the residual of an exponential-form surrogate that upper-bounds the true likelihood.

Corollary 3.1. *Suppose there exist basis functions $\phi(\theta)$, parameter λ , and a constant $\kappa(D_C)$ such that, for all θ , $\lambda^\top \phi(\theta) + \kappa(D_C) \geq \log p(D_C | \theta)$. Under the conditions of Thm. 3.1, the maximal variational family induced by the surrogate is*

$$\mathcal{Q}_c = \{q \in \Delta : \mathbb{E}_q[r(\theta)] \geq \bar{r}\},$$

where the residual is $r(\theta) := \lambda^\top \phi(\theta) + \kappa(D_C) - \log p(D_C | \theta)$ and $\bar{r} := \mathbb{E}_{\hat{q}_i}[r(\theta)]$ is a constant independent of i .

This expression admits a simple interpretation: if $\hat{q}_i$ concentrates on regions where the residual $r(\theta)$ is small, then $\bar{r}$ approaches 0, and $\mathcal{Q}_c$ becomes closer to the full space Δ .

The second corollary identifies a favorable regime in which $\mathcal{Q}_c$ coincides with the full space.

Corollary 3.2. *Assume all priors $p_i \in \mathcal{P}$ belong to a common exponential family and, for the dataset D_C , the*

Bayesian posterior under each p_i also remains within that family. Then the maximal variational family satisfies $\mathcal{Q}_c = \Delta$.

This corollary provides an idealized setting in which a reusable task-update factor always exists. Detailed proofs of the theorem and corollaries see App. C.

3.2. Bayesian Task Update Transfer (BTransfer)

A practical method for transferring fine-tuning knowledge benefits from the existence of a reusable task-update factor $\mathcal{U}(\theta; D_C)$ that is both invariant across base models and admits a fixed-dimensional parameterization. Building on the theoretical results in the previous section, an ideal regime occurs when the prior and the corresponding Bayesian posterior both lie in a shared exponential family. In this case, the maximal admissible variational family coincides with the full space, so the variational problem in Eq. (1) becomes unconstrained. This matches standard fine-tuning practice: we typically do not impose explicit structural restrictions on the variational family. Therefore, we develop our transfer method within this idealized setting first.

Our core insight is as follows. If we can extract the reusable update factor $\mathcal{U}(\theta; D_C)$ w.r.t. dataset D_C from a single fine-tuning run, then, when presented with a new pre-trained model (a new prior), we do not need to fine-tune the model again. Instead, we reapply the same multiplicative update to the new prior to obtain resulting posterior, which matches the Bayesian posterior that would be obtained by fine-tuning using D_C .

Consider a collection of priors $\mathcal{P} = \{p_i\}_{i=1}^N$. Assume that each prior and its Bayesian posterior under the same dataset D_C belong to a common exponential family, sharing the same sufficient statistics $T(\theta)$, base measure $h(\theta)$ and log-partition function $A(\cdot)$ (Blei et al., 2017):

$$\begin{aligned} p_i(\theta) &= h(\theta) \exp\left(\eta_i^\top T(\theta) - A(\eta_i)\right) \\ p_i(\theta | D_C) &= h(\theta) \exp\left((\eta_i^C)^\top T(\theta) - A(\eta_i^C)\right), \end{aligned} \quad (2)$$

where η_i and η_i^C are the natural parameters of the prior and posterior, respectively. By Cor. 3.2, there exists a task-update factor $\mathcal{U}(\theta; D_C) = \exp(\lambda^\top \phi(\theta))$ such that, for every $p_i \in \mathcal{P}$, $q_i(\theta) \propto \mathcal{U}(\theta; D_C)p_i(\theta)$. Equivalently, the posterior-to-prior ratio is shared across all priors:

$$\begin{aligned} \mathcal{U}(\theta; D_C) &\propto \frac{p_1(\theta | D_C)}{p_1(\theta)} = \dots = \frac{p_N(\theta | D_C)}{p_N(\theta)} \\ &= \frac{h(\theta) \exp\left((\eta_i^C)^\top T(\theta) - A(\eta_i^C)\right)}{h(\theta) \exp\left(\eta_i^\top T(\theta) - A(\eta_i)\right)} \\ &\propto \exp\left((\eta_i^C - \eta_i)^\top T(\theta)\right), \end{aligned} \quad (3)$$

where “ $\propto$ ” absorbs the θ -independent normalizing constant $A(\eta_i^C) - A(\eta_i)$.

Thus, a single fine-tuning run suffices to estimate the reusable update. In other words, if we have a certain prior and corresponding Bayesian posterior, we may set $\phi(\theta) = T(\theta)$ and $\lambda = \eta_i^C - \eta_i$, which yields the reusable task-update factor $\mathcal{U}(\theta; D_C) \propto \exp((\eta_i^C - \eta_i)^\top T(\theta))$. Then, we can apply $\mathcal{U}(\theta; D_C)$ to other prior $p_j(\theta)$:

$$\mathcal{U}(\theta; D_C)p_j(\theta) \propto h(\theta) \exp((\eta_j + (\eta_i^C - \eta_i))^\top T(\theta)),$$

which recovers the exact Bayesian posterior $p_j(\theta | D_C)$ up to the normalizing constant.

3.3. Instantiating BTransfer for Deep Networks

Although BTransfer is developed in the shared exponential-family setting, we now describe how to instantiate it for deep networks. Concretely, we implement BTransfer via a three-step “lift–transfer–return” procedure: 1) lift model parameters to distributions using a Laplace approximation (Daxberger et al., 2021); 2) transfer the recovered task-update factor in an exponential-family distribution space; and 3) return by mapping the updated posterior distribution back to parameter space.

We consider a minimal setup. Let θ_{old} be an old base model pre-trained on $D_{\text{old}}^{\text{pre}}$, and let θ_{old}^C denote the checkpoint obtained by fine-tuning θ_{old} on the task dataset D_C . Our goal is to transfer the fine-tuning knowledge to a new base model θ_{new} pre-trained on $D_{\text{new}}^{\text{pre}}$.

Lift. For each of the three models—treated as point estimates and denoted uniformly by θ_m with $m \in \{\text{old}, \text{old}^C, \text{new}\}$, we lift θ_m to a distribution via a Laplace approximation. Specifically, we approximate the local distribution around θ_m by a Gaussian written in exponential-family form as:

$$\pi_m(\theta) \propto \exp\left([b_m; \text{vec}(-\frac{1}{2}\Lambda_m)]^\top [\theta; \text{vec}(\theta\theta^\top)]\right), \quad (4)$$

where $[\cdot; \cdot]$ denotes column-wise concatenation, $b_m = \Lambda_m \theta_m \in \mathbb{R}^m$, and Λ_m is the precision matrix, given by the Hessian of the training objective at θ_m . In practice, computing the full Hessian is expensive for deep networks, so we follow Yun & Liu (2022) and approximate Λ_m using a diagonal Fisher information matrix. Note that θ_{old}^C is a mode of the objective that combines the task-data likelihood with the pre-training–induced regularizer; accordingly, Λ_{old}^C should be estimated using $D_{\text{old}}^{\text{pre}} \cup D_C$. Comparing with Eq. (2), $\pi_m(\theta)$ corresponds to an exponential-family parameterization with natural parameter $\eta_m = [\Lambda_m \theta_m; \text{vec}(-\frac{1}{2}\Lambda_m)]$, sufficient statistic $T(\theta) = [\theta; \text{vec}(\theta\theta^\top)]$, base measure $h(\theta) \equiv 1$, and all θ -independent terms absorbed into the normalizing constant. See App. D for more details.

Algorithm 1 BTransfer

Input: model parameters $\theta_{\text{old}}, \theta_{\text{old}}^C, \theta_{\text{new}}$, datasets $D_C, D_{\text{old}}^{\text{pre}}, D_{\text{new}}^{\text{pre}}$.

Diagonal empirical Fisher:

$$\Lambda_{\text{old}} \leftarrow \text{FISHER}(\theta_{\text{old}}, D_{\text{old}}^{\text{pre}})$$

$$\Lambda_{\text{old}}^C \leftarrow \text{FISHER}(\theta_{\text{old}}^C, D_C \cup D_{\text{old}}^{\text{pre}})$$

$$\Lambda_{\text{new}} \leftarrow \text{FISHER}(\theta_{\text{new}}, D_{\text{new}}^{\text{pre}}) \quad \{\text{FISHER}(\theta, D) \text{ denotes the diagonal Fisher information matrix evaluated at } \theta \text{ on dataset } D.\}$$

Parameter update:

$$\theta_{\text{new}}^C \leftarrow \text{UPDATE}(\Lambda_{\text{old}}, \Lambda_{\text{old}}^C, \Lambda_{\text{new}}) \quad \{\text{Defined in Eq. (5).}\}$$

Output: transferred parameters θ_{new}^C

Transfer. Now, we can identify the reusable task-update factor $\mathcal{U}(\theta; D_C)$ and use it to obtain the updated posterior of the new model. By Eq. (3),

$$\begin{aligned} \mathcal{U}(\theta; D_C) &\propto \pi_{\text{old}}^C(\theta) / \pi_{\text{old}}(\theta) \propto \exp((\eta_{\text{old}}^C - \eta_{\text{old}})^\top T(\theta)) \\ &= \exp\left([\Delta b; \text{vec}(-\frac{1}{2}\Delta\Lambda)]^\top T(\theta)\right), \end{aligned}$$

where $\Delta\Lambda = \Lambda_{\text{old}}^C - \Lambda_{\text{old}}$ and $\Delta b = b_{\text{old}}^C - b_{\text{old}}$. It is worth noting that this update factor can be computed once after fine-tuning the old model on D_C and stored for later reuse. Applying $\mathcal{U}(\theta; D_C)$ to $\pi_{\text{new}}(\theta)$ yields the updated posterior for the new model:

$$\begin{aligned} &\pi_{\text{new}}^C(\theta) \\ &\propto \mathcal{U}(\theta; D_C) \pi_{\text{new}}(\theta) \\ &\propto \exp\left([b_{\text{new}} + \Delta b; \text{vec}(-\frac{1}{2}(\Lambda_{\text{new}} + \Delta\Lambda))]^\top T(\theta)\right). \end{aligned}$$

Thus, the natural parameters of the new posterior are updated as $b_{\text{new}}^C = b_{\text{new}} + \Delta b$ and $\Lambda_{\text{new}}^C = \Lambda_{\text{new}} + \Delta\Lambda$.

Return. Since the natural parameters satisfy $b_{\text{new}}^C = \Lambda_{\text{new}}^C \theta_{\text{new}}^C$, we map this updated posterior back to a parameter estimate via

$$\begin{aligned} &\theta_{\text{new}}^C \\ &= (\Lambda_{\text{new}}^C)^{-1} b_{\text{new}}^C \\ &= (\Lambda_{\text{new}} + \Delta\Lambda)^{-1} (b_{\text{new}} + \Delta b) \\ &= (\Lambda_{\text{new}} + (\Lambda_{\text{old}}^C - \Lambda_{\text{old}}))^{-1} (\Lambda_{\text{new}} \theta_{\text{new}} + (\Lambda_{\text{old}}^C \theta_{\text{old}}^C - \Lambda_{\text{old}} \theta_{\text{old}})). \end{aligned} \quad (5)$$

Intuitively, BTransfer estimates the precision matrices Λ at $\theta_{\text{old}}, \theta_{\text{old}}^C$, and θ_{new} using a diagonal Fisher approximation, and then transfers the fine-tuning knowledge to θ_{new} through Eq. (5). In the shared exponential-family regime, this procedure recovers the exact Bayesian posterior; for deep networks, it provides a practical approximation. Algorithm 1 summarizes the workflow.

Table 1. Main results transferring fine-tuning knowledge from roberta-base to legal/mental. We report accuracy (%) [↑] for non-transfer baselines (θ_{old}^C , θ_{new} , θ_{new}^C), parameter-space baselines (Param Δ , TransFusion), and BTransfer (ours). **Bold** indicates the best result among transfer methods.

Dataset	SST-2		MNLI		QNLI		RTE		Subj		MPQA		MR	
Model	legal	mental	legal	mental	legal	mental	legal	mental	legal	mental	legal	mental	legal	mental
θ_{new}	56.31	62.96	41.05	47.51	48.82	49.88	50.90	57.04	54.80	52.55	50.50	52.55	55.90	59.10
θ_{old}^C	90.83	90.83	70.61	70.61	76.40	76.40	64.26	64.26	93.05	93.05	84.75	84.75	84.35	84.35
θ_{new}^C	89.91	91.17	72.69	70.12	81.42	78.11	63.54	68.59	90.20	94.25	83.25	85.65	82.90	88.30
Param Δ	82.57	90.48	47.17	61.29	51.36	61.18	51.62	63.90	87.15	92.85	78.15	75.40	79.20	85.10
TransFusion	82.57	90.48	47.21	61.23	51.36	61.03	51.62	63.90	87.15	92.85	78.05	74.70	79.20	85.00
BTransfer	91.17	90.60	70.32	70.34	75.18	76.24	63.54	63.54	91.65	91.55	83.15	84.30	84.30	83.95

Table 2. Main results transferring fine-tuning knowledge from roberta-base to codebert/4m. We report accuracy (%) [↑] for non-transfer baselines (θ_{old}^C , θ_{new} , θ_{new}^C), parameter-space baselines (Param Δ , TransFusion), and BTransfer (ours). **Bold** indicates the best result among transfer methods.

Dataset	SST-2		MNLI		QNLI		RTE		Subj		MPQA		MR	
Model	codebert	4m	codebert	4m	codebert	4m	codebert	4m	codebert	4m	codebert	4m	codebert	4m
θ_{new}	55.05	51.03	35.91	40.02	49.37	49.61	51.99	51.26	50.90	50.30	57.80	50.00	56.25	50.00
θ_{old}^C	90.83	90.83	70.61	70.61	76.40	76.40	64.26	64.26	93.05	93.05	84.75	84.75	84.35	84.35
θ_{new}^C	85.21	88.07	68.75	71.04	80.32	81.37	61.73	66.43	90.25	93.20	81.85	85.20	79.55	84.25
Param Δ	70.76	60.21	35.44	51.01	50.59	50.91	47.29	51.99	63.20	78.25	65.75	77.35	70.65	65.60
TransFusion	70.87	59.75	35.44	50.85	50.59	50.91	47.29	51.99	63.15	78.30	65.50	77.20	70.80	65.45
BTransfer	90.83	90.94	70.97	70.37	77.27	77.19	65.34	66.06	86.45	92.60	84.40	84.30	82.05	84.65

4. Experiments

4.1. Experimental Setup

Models. We use RoBERTa (Liu et al., 2019) models with different pre-trained checkpoints as base models. Specifically, we use roberta-base as the old pre-trained model θ_{old} and treat several domain-adapted RoBERTa variants as new pre-trained models θ_{new} , including legal (legal-roberta-base) (Geng et al., 2021), mental (mental-roberta-base) (Ji et al., 2022), codebert (codebert-base-mlm) (Feng, 2020), and 4m (radbert-roberta-4m) (Yan et al., 2022). All pre-trained checkpoints are loaded from publicly available weights on the Hugging Face Hub. In addition, we directly use the publicly released fine-tuned roberta-base checkpoints from Panigrahi et al. (2023), which are trained on the datasets described below, and treat them as the old fine-tuned models θ_{old}^C in our experiments.

Datasets. We evaluate the proposed method’s knowledge-transfer capability on a subset of the GLUE benchmark (Wang et al., 2018), including SST-2 (Socher et al., 2013), MNLI (Williams et al., 2018), RTE, and QNLI (Rajpurkar et al., 2016). We additionally consider three widely used sentence classification datasets: MR, MPQA, and Subj. Following Gao et al. (2021), we implemented all tasks as **prompt-based classification**, which reformulates classification as masked language modeling via a task-specific template and label-word mapping. For Fisher estimation, we sample from the English Wikipedia text (20220301 dump) to compute Fisher matrices at θ_{old} and θ_{new} , and use the

training sets of the above downstream datasets to estimate the Fisher at each fine-tuned checkpoint θ_{old}^C .

Baselines. We compare BTransfer against several representative parameter-space transfer methods. Param Δ (Cao et al., 2025) transfers knowledge by adding the parameter difference between the old fine-tuned model and its pre-trained counterpart, $\Delta\theta = \theta_{old}^C - \theta_{old}$, to the new pre-trained model, i.e., $\theta_{new}^C = \theta_{new} + \Delta\theta$. In contrast, TransFusion (Rinaldi et al., 2025a) introduces an additional permutation step that maps $\Delta\theta$ into a low-loss basin aligned with the new model’s loss landscape before applying it. As non-transfer reference baselines, we also report: (i) the old fine-tuned model θ_{old}^C obtained by fine-tuning θ_{old} on D_C , (ii) the new pre-trained model θ_{new} without transfer, and (iii) the new fine-tuned model θ_{new}^C , obtained by directly fine-tuning θ_{new} on D_C following the framework of Panigrahi et al. (2023), with batch size 8, learning rate 5×10^{-4} , and $16 \times (\text{number of classes}) \times 512$ training iterations. For BTransfer, we estimate the Fisher matrix using 5,000 samples by default.

4.2. Experimental Results

Baseline Comparisons. Tabs. 1 and 2 report the main results. Averaged over all tasks and all models, BTransfer achieves an accuracy of about 80.1%, which is within 0.5 points of the old fine-tuned model θ_{old}^C ($\approx 80.6\%$), while exceeding the new pre-trained model θ_{new} ($\approx 51.4\%$) by nearly 30 points.

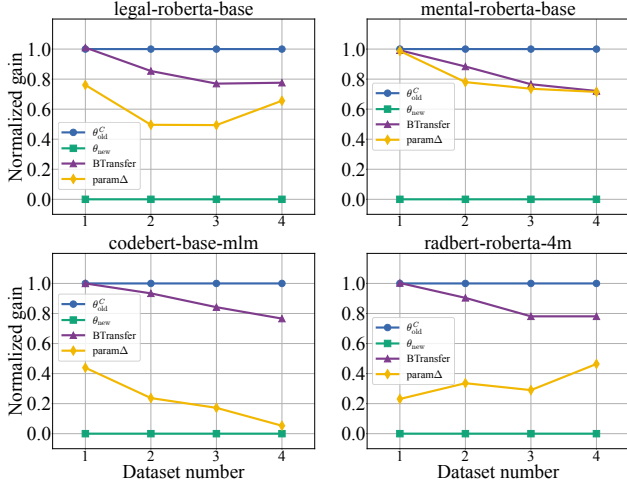


Figure 2. Normalized gain for joint transfer of multiple task updates.

Compared with parameter-space methods, BTransfer yields robust improvements. On the legal and mental backbones, Param Δ attains roughly 72% accuracy on average and TransFusion performs similarly, whereas BTransfer reaches about 80%. The advantage is even more pronounced on codebert and 4m: Param Δ and TransFusion achieve average accuracies of 59.93% and 59.86%, while BTransfer reaches 80.24%. In some cases, parameter-space methods even underperform θ_{new} (e.g., on MNLI and RTE with codebert). By contrast, BTransfer matches θ_{old}^C and occasionally exceeds it (e.g., on QNLI and RTE). Overall, BTransfer consistently outperforms parameter-space transfer baselines and substantially close to the performance of the fine-tuned model.

Joint Transfer of Multiple Task Updates. Beyond transferring a single fine-tuning update, we also investigate whether BTransfer can transfer multiple task updates in a single step. Specifically, we constructed a series of task sets, including {SST-2}, {SST-2, MNLI}, {SST-2, MNLI, QNLI}, {SST-2, MNLI, QNLI, RTE}. For each task $t \in T$, we compute its natural-parameter increments ($\Delta\Lambda_t, \Delta b_t$). For joint transfer over a set T , we aggregate these increments by summation,

$$\Delta\Lambda_T = \sum_{t \in T} \Delta\Lambda_t, \quad \Delta b_T = \sum_{t \in T} \Delta b_t,$$

and then plug $(\Delta\Lambda_T, \Delta b_T)$ into Eq. (5) to obtain the jointly updated parameters θ_{new}^C for the new model. We include Param Δ as a parameter-space baseline, which sums task vectors $\Delta\theta_t$ and applies $\Delta\theta_T = \sum_{t \in T} \Delta\theta_t$ directly to θ_{new} .

We evaluate performance using a normalized gain metric that maps the new pre-trained model θ_{new} to 0 and

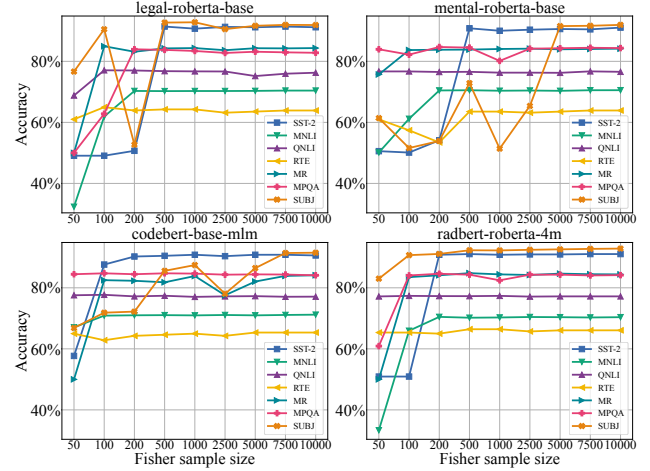


Figure 3. Effect of Fisher sample size on BTransfer performance.

the old fine-tuned model θ_{old}^C to 1. A normalized gain > 1 indicates outperforming θ_{old}^C , while a gain < 0 indicates underperforming θ_{new} (see App. E.3 for details). Fig. 2 shows that BTransfer nearly matches θ_{old}^C in the single-task setting and degrades smoothly as more tasks are transferred jointly, whereas Param Δ exhibits less stable degradation. Across all task sets and model settings, BTransfer consistently outperforms Param Δ . In the 4-task case, BTransfer achieves a mean normalized gain of 0.761, substantially higher than the 0.473 achieved by Param Δ . Overall, BTransfer enables joint transfer of multiple fine-tuning updates in a single operation while preserving most of the model performance.

Sensitivity Analysis. We next examine how sensitive BTransfer is to the number of samples used to estimate the Fisher information matrix. For each new pre-trained model, we vary the Fisher sample size over {50, 100, 200, 500, 1000, 2,500, 5,000, 7,500, 10,000} and evaluate BTransfer on all seven downstream tasks. The resulting accuracy curves are shown in Fig. 3.

Across nearly all models and tasks, performance increases rapidly as the sample size grows from 50 to roughly 500, and then quickly saturates. For example, on the legal and 4m backbones, each task reaches near-optimal accuracy with only a few hundred samples; increasing the sample size from 1,000 to 10,000 yields only marginal gains. The mental and codebert backbones show a similar trend: for most tasks, the curves begin to flatten around 500 samples and remain stable at larger sample sizes.

Overall, these results suggest that BTransfer is not overly sensitive to the Fisher sample size. Estimating the diagonal Fisher matrix with only a few hundred samples drawn from

Table 3. Main results transferring fine-tuning knowledge from gemma-2-2b-it to BG-2B/TiTU-2B. We report accuracy (%) $\uparrow$ for non-transfer baselines (θ_{old}^C , θ_{new}), parameter-space baseline (Param Δ), and BTransfer (ours).

Dataset	Coding		Safety		Math		Multilingual		Instruction	
Model	BG-2B	TiTU-2B	BG-2B	TiTU-2B	BG-2B	TiTU-2B	BG-2B	TiTU-2B	BG-2B	TiTU-2B
θ_{new}	24.89	10.96	68.65	42.91	50.72	12.43	37.31	31.52	22.37	3.51
θ_{old}^C	41.90	41.90	71.10	71.10	63.21	63.21	40.86	40.86	41.64	41.64
Param Δ	18.05	9.20	47.36	34.83	44.66	15.85	34.98	31.62	23.48	23.48
BTransfer	32.45	28.07	72.59	46.18	56.26	31.84	41.78	40.54	34.01	33.27

Table 4. Main results transferring fine-tuning knowledge from gemma-2-2b-it to Atlas-2B/MG2-2B. We report accuracy (%) $\uparrow$ for non-transfer baselines (θ_{old}^C , θ_{new}), parameter-space baseline (Param Δ), and BTransfer (ours).

Dataset	Coding		Safety		Math		Multilingual		Instruction	
Model	Atlas-2B	MG2-2B	Atlas-2B	MG2-2B	Atlas-2B	MG2-2B	Atlas-2B	MG2-2B	Atlas-2B	MG2-2B
θ_{new}	34.24	38.17	75.27	65.57	41.02	42.38	39.85	39.68	39.74	35.86
θ_{old}^C	41.90	41.90	71.10	71.10	63.21	63.21	40.86	40.86	41.64	41.64
Param Δ	34.03	38.73	49.64	42.29	52.62	55.72	38.19	39.81	37.15	37.52
BTransfer	37.11	39.42	81.15	75.86	47.08	52.09	41.16	41.1	42.14	41.96

the available training corpus already achieves near-optimal performance. In this regime, the method remains stable over a broad range of sample sizes. This supports the practicality of BTransfer in settings where only a small subset of training examples can be afforded for Fisher estimation due to computational budget constraints.

4.3. Large Language Model Transfer

Experimental setup. We evaluate BTransfer in the large language model setting for transferring fine-tuning knowledge. We use gemma-2-2b-it³ as the old pre-trained model θ_{old} , which is an instruction-tuned variant of Gemma2-2B (Team et al., 2024). We then transfer the fine-tuning knowledge learned by θ_{old}^C to several new pre-trained models θ_{new} , including BG-2B (BgGPT-Gemma-2-2.6B-IT-v1.0⁴), TiTU-2B (titulm-gemma-2-2b⁵), Atlas-2B (atlas-chat-2b) (Shang et al., 2025), and MG2-2B (Meditron3-Gemma2-2B⁶). We initialize θ_{old} and each θ_{new} from publicly released checkpoints on the Hugging Face Hub, and initialize θ_{old}^C from the checkpoints provided by He et al. (2025). For the baselines, we do not include TransFusion in these experiments because it assumes a shared Q/K/V head partition to define consistent head-wise permutations, whereas Gemma2-2B uses Grouped-Query Attention with different numbers of Q and K/V heads, making TransFusion not directly applicable.

We evaluate BTransfer using the protocol introduced

by He et al. (2025). Specifically, we consider five capability categories: Coding, Safety, Math, Multilingual, and Instruction-following. For each category C , we treat the model obtained by fine-tuning θ_{old} on the corresponding training dataset as the old fine-tuned model θ_{old}^C . To evaluate transfer, we estimate the Fisher matrix using a subset of the category’s training set and assess the transferred models on the designated evaluation sets. When a category contains multiple evaluation datasets, we report the arithmetic mean of the corresponding metrics across those datasets. Detailed settings for datasets see App. E.4.

Main results. Tabs. 3 and 4 report the main results. Compared to the new pre-trained model θ_{new} , BTransfer achieves average gains of +6.63, +15.71, +3.70, and +5.75 points on BG-2B, TiTU-2B, Atlas-2B, and MG2-2B, respectively. In contrast, the parameter-space baseline Param Δ is unstable and often degrades performance on average (e.g., -7.08, -3.70, and -1.52 points on BG-2B, Atlas-2B, and MG2-2B), despite bringing limited gains on TiTU-2B.

Across tasks, BTransfer shows particular effectiveness in safety and multilingual tasks. It achieves high scores across several new bases, even outperforming the old fine-tuned model θ_{old}^C , including 72.59% on BG-2B, 81.15% on Atlas-2B, and 75.86% on MG2-2B. It also consistently improves multilingual performance to approximately 41% on BG-2B, Atlas-2B, and MG2-2B. Moreover, BTransfer substantially boosts instruction-following on weaker models, such as TiTU-2B (3.51% $\rightarrow$ 33.27%). Overall, these results support BTransfer as an effective approach for robust fine-tuning knowledge transfer across instruction-tuned variants.

³google/gemma-2-2b-it

⁴INSAIT-Institute/BgGPT-Gemma-2-2.6B-IT-v1.0

⁵hishab/titulm-gemma-2-2b-v1.1

⁶OpenMeditron/Meditron3-Gemma2-2B

Table 5. Main results transferring fine-tuning knowledge from `commonpool_l1slb_b8k` to `datacomp/laion`. We report accuracy (%) [$\uparrow$] for non-transfer baselines ($\theta_{old}^C, \theta_{new}$), parameter-space baselines (Param Δ , TransFusion), and BTransfer (ours).

Dataset	SVHN		GTSRB		EuroSAT		DTD	
Model	datacomp	laion	datacomp	laion	datacomp	laion	datacomp	laion
θ_{new}	46.00	42.95	43.33	33.18	49.15	41.04	47.50	41.91
θ_{old}^C	67.85	67.85	99.33	99.33	98.81	98.81	75.00	75.00
Param Δ	23.92	27.19	38.07	30.38	41.32	38.89	47.39	41.81
TransFusion	46.37	43.86	45.11	34.24	50.98	45.02	46.38	41.91
BTransfer	47.84	50.98	53.48	49.38	50.25	48.59	53.62	46.65

4.4. Vision Model Transfer

Experimental setup. Beyond NLP tasks, we evaluate the fine-tuning transfer performance of BTransfer in the vision domain using CLIP ViT-B/16 (Radford et al., 2021; Cherti et al., 2023). We use `commonpool_l1slb_b8k` as the old pre-trained model θ_{old} , and consider two new pre-trained models, `datacomp` (`datacomp_l1slb_b8k`) and `laion` (`commonpool_l1laion_slb_b8k`), as θ_{new} . All checkpoints are released by Gadre et al. (2023). We evaluate transfer performance on four image classification benchmarks: SVHN (Netzer et al., 2011), GTSRB (Stal Kamp et al., 2011), EuroSAT (Helber et al., 2019), and DTD (Cimpoi et al., 2014).

Main results. As shown in Tab. 5, BTransfer achieves the best or second-best performance across all vision transfer settings. When transferring to `datacomp`, BTransfer improves over the new pre-trained model θ_{new} by +1.84, +10.15, +1.10, and +6.12 points on SVHN, GTSRB, EuroSAT, and DTD, respectively. The gains are larger when transferring to `laion`, with improvements of +8.03, +16.20, +7.55, and +4.74 points across the same four tasks. In contrast, Param Δ often degrades performance below θ_{new} . TransFusion is competitive in some cases, but BTransfer delivers stronger overall performance. Overall, these results support the effectiveness of BTransfer in the vision domain.

5. Conclusion

In this work, we cast fine-tuning as a variational inference problem and formalize the notion of a reusable task-update factor via two criteria: invariance across base models and a fixed-dimensional parameterization. Our theoretical results identify a sharp boundary for reusability: such factors exist when the variational family is a half-space, and this choice is maximal admissible among convex families. Moreover, the shared exponential-family setting further provides an ideal regime in which reusable factors arise naturally. These insights motivate BTransfer, which extracts a task-update factor from a single fine-tuning run on an old model and applies it to a new prior to obtain the corresponding Bayesian

posterior. We further propose a “lift–transfer–return” procedure that makes BTransfer applicable to deep networks. Extensive experiments demonstrate that BTransfer can effectively reuse fine-tuning knowledge across evolving model families, providing a theoretically grounded alternative to existing parameter-space transfer methods. More discussion is provided in Apps. A and B.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 62072485), the Guangdong Provincial Key Area S&T Program (No. 2025B0909030001), and the Guangdong Basic and Applied Basic Research Foundation (No. 2025A1515010248). We also thank Yaming Guo for helpful discussion.

Impact Statement

This paper formalizes reusable task-update factors for fine-tuning transfer and establishes structural existence results, motivating BTransfer as an effective method for reusing fine-tuning knowledge across models without retraining. Given the theoretical nature of our study, we have not identified any direct ethical concerns or immediate negative societal impacts specific to our framework. However, as with related advances, the techniques developed here could be misused to propagate undesirable behaviors or biases across models if the underlying fine-tuning updates are themselves misaligned. Appropriate safeguards, monitoring, and governance are therefore important to ensure responsible use in deployment. Thus, the societal impact of this work is largely indirect and depends on how practitioners apply these techniques in real-world systems.

References

- Ainsworth, S., Hayase, J., and Srinivasa, S. Git re-basin: Merging models modulo permutation symmetries. In *International Conference on Learning Representations*, 2023.
- Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. Varia-

- tional inference: A review for statisticians. *Journal of the American statistical Association*, 112(518):859–877, 2017.
- Borse, S., Kadambi, S., Pandey, N. P., Bhardwaj, K., Ganapathy, V., Priyadarshi, S., Garrepalli, R., Esteves, R., Hayat, M., and Porikli, F. Foura: Fourier low-rank adaptation. In *Advances in Neural Information Processing Systems*, volume 37, pp. 71504–71539, 2024.
- Cai, R., Dong, Z., Dong, J., and Wang, X. Dynamic adapter with semantics disentangling for cross-lingual cross-modal retrieval. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 1907–1916, 2025.
- Cao, S., Wu, M., Prasad, K., Tian, Y., and Liu, Z. Param Δ for direct mixing: Post-train large language model at zero cost. In *International Conference on Learning Representations*, 2025.
- Chalkidis, I., Garneau, N., Goanta, C., Katz, D., and Søgaard, A. LeXFiles and LegalLAMA: Facilitating English multinational legal language model development. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15513–15535, July 2023.
- Chen, X., Wang, X., Changpinyo, S., Piergiovanni, A., Padlewski, P., Salz, D., Goodman, S., Grycner, A., Mustafa, B., Beyer, L., Kolesnikov, A., Puigcerver, J., Ding, N., Rong, K., Akbari, H., Mishra, G., Xue, L., Thapliyal, A. V., Bradbury, J., Kuo, W., Seyedhosseini, M., Jia, C., Ayan, B. K., Ruiz, C. R., Steiner, A. P., Angelova, A., Zhai, X., Houlsby, N., and Soricut, R. PaLI: A jointly-scaled multilingual language-image model. In *the Eleventh International Conference on Learning Representations*, 2023.
- Cherti, M., Beaumont, R., Wightman, R., Wortsman, M., Ilharco, G., Gordon, C., Schuhmann, C., Schmidt, L., and Jitsev, J. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2818–2829, 2023.
- Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tai, Y., Fedus, W., Li, Y., Wang, X., Dehghani, M., Brahma, S., Webson, A., Gu, S. S., Dai, Z., Suzgun, M., Chen, X., Chowdhery, A., Castro-Ros, A., Pellat, M., Robinson, K., Valter, D., Narang, S., Mishra, G., Yu, A., Zhao, V., Huang, Y., Dai, A., Yu, H., Petrov, S., Chi, E. H., Dean, J., Devlin, J., Roberts, A., Zhou, D., Le, Q. V., and Wei, J. Scaling instruction-finetuned language models. *J. Mach. Learn. Res.*, 25(1), 2024. ISSN 1532-4435.
- Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. Describing textures in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3606–3613, 2014.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Crisostomi, D., Fumero, M., Baieri, D., Bernard, F., and Rodolà, E. c^2m^3 : Cycle-consistent multi-model merging. In *Advances in Neural Information Processing Systems*, 2024.
- Daxberger, E., Kristiadi, A., Immer, A., Eschenhagen, R., Bauer, M., and Hennig, P. Laplace redux - effortless bayesian deep learning. In *Advances in Neural Information Processing Systems*, 2021.
- Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.-M., Chen, W., et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature machine intelligence*, 5(3):220–235, 2023.
- Feng, Z. Codebert: A pre-trained model for program-ming and natural languages. *arXiv preprint arXiv:2002.08155*, 2020.
- Fu, Z., Yang, H., So, A. M.-C., Lam, W., Bing, L., and Collier, N. On the effectiveness of parameter-efficient fine-tuning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pp. 12799–12807, 2023.
- Gadre, S. Y., Ilharco, G., Fang, A., Hayase, J., Smyrnis, G., Nguyen, T., Marten, R., Wortsman, M., Ghosh, D., Zhang, J., et al. Datacomp: In search of the next generation of multimodal datasets. In *Advances in Neural Information Processing Systems*, volume 36, pp. 27092–27112, 2023.
- Gao, T., Fisch, A., and Chen, D. Making pre-trained language models better few-shot learners. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers)*, pp. 3816–3830, 2021.
- Gao, Z., Wang, Q., Chen, A., Liu, Z., Wu, B., Chen, L., and Li, J. Parameter-efficient fine-tuning with discrete fourier transform. In *International Conference on Machine Learning*, 2024.
- Gargiulo, A. A., Crisostomi, D., Bucarelli, M. S., Scardapane, S., Silvestri, F., and Rodola, E. Task singular vectors: Reducing task interference in model merging. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 18695–18705, 2025.

- Geng, S., Lebre, R., and Aberer, K. Legal transformer models may not always help. *arXiv preprint arXiv:2109.06862*, 2021.
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Gu, N., Fu, P., Liu, X., Ma, K., Lin, Z., and Wang, W. Adapt once, thrive with updates: Transferable parameter-efficient fine-tuning on evolving base models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 14765–14783, 2025.
- Guerrero-Peña, F. A., Medeiros, H. R., Dubail, T., Aminbeidokhti, M., Granger, E., and Pedersoli, M. Re-basin via implicit sinkhorn differentiation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 20237–20246, 2023.
- Guo, H., Yang, J., Liu, J., Bai, J., Wang, B., Li, Z., Zheng, T., Zhang, B., Peng, J., and Tian, Q. Logformer: A pre-train and tuning pipeline for log anomaly detection. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pp. 135–143, 2024.
- Han, S., Rao, K., Ettinger, A., Jiang, L., Lin, B. Y., Lambert, N., Choi, Y., and Dziri, N. Wildguard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of LLMs. In *Advances in Neural Information Processing Systems*, 2024.
- Hayou, S., Ghosh, N., and Yu, B. LoRA+: Efficient low rank adaptation of large models. In *International Conference on Machine Learning*, 2024.
- He, R., Liu, L., Ye, H., Tan, Q., Ding, B., Cheng, L., Low, J., Bing, L., and Si, L. On the effectiveness of adapter-based tuning for pretrained language model adaptation. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers)*, pp. 2208–2222, 2021.
- He, Y., Zeng, S., Hu, Y., Yang, R., Zhang, T., and Zhao, H. Mergebench: A benchmark for merging domain-specialized LLMs. In *Advances in Neural Information Processing Systems*, 2025.
- Helber, P., Bischke, B., Dengel, A., and Borth, D. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.
- Houlsby, N., Giurghi, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. Parameter-efficient transfer learning for nlp. In *International Conference on Machine Learning*, pp. 2790–2799, 2019.
- Howard, J. and Ruder, S. Universal language model fine-tuning for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 328–339, July 2018.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, volume 1, pp. 3, 2022.
- Huang, C., Ye, P., Chen, T., He, T., Yue, X., and Ouyang, W. EMR-merging: Tuning-free high-performance model merging. In *Advances in Neural Information Processing Systems*, 2024.
- Huang, S., Dong, L., Wang, W., Hao, Y., Singhal, S., Ma, S., Lv, T., Cui, L., Mohammed, O. K., Patra, B., Liu, Q., Aggarwal, K., Chi, Z., Bjorck, J., Chaudhary, V., Som, S., Song, X., and Wei, F. Language is not all you need: Aligning perception with language models. In *Advances in Neural Information Processing Systems*, 2023.
- Ilharco, G., Ribeiro, M. T., Wortsman, M., Schmidt, L., Hajishirzi, H., and Farhadi, A. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.
- Imfeld, M., Galdi, J., Giordano, M., Hofmann, T., Anagnostidis, S., and Singh, S. P. Transformer fusion with optimal transport. In *International Conference on Learning Representations*, 2024.
- Iurada, L., Ciccone, M., and Tommasi, T. Efficient model editing with task-localized sparse fine-tuning. In *International Conference on Learning Representations*, 2025.
- Ji, S., Zhang, T., Ansari, L., Fu, J., Tiwari, P., and Cambria, E. MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare. In *Proceedings of LREC*, 2022.
- Jin, X., Ren, X., Preotiuc-Pietro, D., and Cheng, P. Data-less knowledge fusion by merging weights of language models. In *International Conference on Learning Representations*, 2023.
- Kim, J., Lee, J. H., Kim, S., Park, J., Yoo, K. M., Kwon, S. J., and Lee, D. Memory-efficient fine-tuning of compressed large language models via sub-4-bit integer quantization. In *Advances in Neural Information Processing Systems*, 2023.

- Lai, V., Nguyen, C., Ngo, N., Nguyen, T., Dernoncourt, F., Rossi, R., and Nguyen, T. Okapi: Instruction-tuned large language models in multiple languages with reinforcement learning from human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 318–327, December 2023.
- Lambert, N., Morrison, J., Pyatkin, V., Huang, S., Ivison, H., Brahman, F., Miranda, L. J. V., Liu, A., Dziri, N., Lyu, X., Gu, Y., Malik, S., Graf, V., Hwang, J. D., Yang, J., Bras, R. L., Tafford, O., Wilhelm, C., Soldaini, L., Smith, N. A., Wang, Y., Dasigi, P., and Hajishirzi, H. Tulu 3: Pushing frontiers in open language model post-training. In *Second Conference on Language Modeling*, 2025.
- Lester, B., Al-Rfou, R., and Constant, N. The power of scale for parameter-efficient prompt tuning. In *Proceedings of Conference on Empirical Methods in Natural Language Processing*, pp. 3045–3059, November 2021.
- Liu, H., Wang, Y., Liu, H., Sun, F., and Yao, A. Small scale data-free knowledge distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6008–6016, 2024a.
- Liu, J., Xia, C. S., Wang, Y., and Zhang, L. Is your code generated by chatgpt really correct? rigorous evaluation of large language models for code generation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 21558–21572, 2023.
- Liu, S.-Y., Wang, C.-Y., Yin, H., Molchanov, P., Wang, Y.-C. F., Cheng, K.-T., and Chen, M.-H. Dora: Weight-decomposed low-rank adaptation. In *International Conference on Machine Learning*, 2024b.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., and Stoyanov, V. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- Matena, M. S. and Raffel, C. Merging models with fisher-weighted averaging. In Oh, A. H., Agarwal, A., Belgrave, D., and Cho, K. (eds.), *Advances in Neural Information Processing Systems*, 2022.
- Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li, N., Basart, S., Li, B., Forsyth, D., and Hendrycks, D. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. In *International Conference on Machine Learning*, 2024.
- Meng, F., Wang, Z., and Zhang, M. Pissa: Principal singular values and singular vectors adaptation of large language models. In *Advances in Neural Information Processing Systems*, volume 37, pp. 121038–121072, 2024.
- Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., Akhtar, N., Barnes, N., and Mian, A. A comprehensive overview of large language models. *ACM Transactions on Intelligent Systems and Technology*, 16(5):1–72, 2025.
- Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. Reading digits in natural images with unsupervised feature learning. In *Neural Information Processing Systems Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.
- Neyshabur, B., Sedghi, H., and Zhang, C. What is being transferred in transfer learning? In *Advances in Neural Information Processing Systems*, 2020.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., and Lowe, R. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022.
- Panigrahi, A., Saunshi, N., Zhao, H., and Arora, S. Task-specific skill localization in fine-tuned language models. In *International Conference on Machine Learning*, pp. 27011–27033, 2023.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pp. 8748–8763, 2021.
- Rajpurkar, P., Zhang, J., Lopyrev, K., and Liang, P. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 2383–2392, November 2016.
- Rinaldi, F., Capitani, G., Bonicelli, L., Crisostomi, D., Bolelli, F., FICARRA, E., Rodolà, E., Calderara, S., and Porrello, A. Update your transformer to the latest release: Re-basin of task vectors. In *International Conference on Machine Learning*, 2025a.
- Rinaldi, F., Panariello, A., Salici, G., Liu, F., Ciccone, M., Porrello, A., and Calderara, S. Gradient-sign masking for task vector transport across pre-trained models. *arXiv preprint arXiv:2510.09658*, 2025b.
- Röttger, P., Kirk, H., Vidgen, B., Attanasio, G., Bianchi, F., and Hovy, D. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational*

- Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 5377–5400, June 2024.
- Shang, G., Abdine, H., Khoubrane, Y., Mohamed, A., Abbahaddou, Y., Ennadir, S., Momayiz, I., Ren, X., Moulines, E., Nakov, P., et al. Atlas-chat: Adapting large language models for low-resource moroccan arabic dialect. In *Proceedings of the First Workshop on Language Models for Low-Resource Languages*, pp. 9–30, 2025.
- Shi, Z. and Lipani, A. Don’t stop pretraining? make prompt-based fine-tuning powerful learner. In *Advances in Neural Information Processing Systems*, volume 36, pp. 5827–5849, 2023.
- Singh, S., Vargus, F., D’souza, D., Karlsson, B. F., Mahendiran, A., Ko, W.-Y., Shandilya, H., Patel, J., Mataciunas, D., O’Mahony, L., Zhang, M., Hettiarachchi, R., Wilson, J., Machado, M., Moura, L., Krzemiński, D., Fadaei, H., Ergun, I., Okoh, I., Alaagib, A., Mudannayake, O., Alyafeai, Z., Chien, V., Ruder, S., Guthikonda, S., Alghamdi, E., Gehrmann, S., Muennighoff, N., Bartolo, M., Kreutzer, J., Üstün, A., Fadaee, M., and Hooker, S. Aya dataset: An open-access collection for multilingual instruction tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11521–11567, August 2024.
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A., and Potts, C. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pp. 1631–1642, October 2013.
- Stallkamp, J., Schlipsing, M., Salmen, J., and Igel, C. The german traffic sign recognition benchmark: a multi-class classification competition. In *The 2011 international joint conference on neural networks*, pp. 1453–1460, 2011.
- Stoica, G., Bolya, D., Bjorner, J. B., Ramesh, P., Hearn, T., and Hoffman, J. Zipit! merging models from different tasks without training. In *International Conference on Learning Representations*, 2024.
- Sung, Y.-L., Cho, J., and Bansal, M. Vl-adapter: Parameter-efficient transfer learning for vision-and-language tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5227–5237, 2022.
- Team, G., Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A. M., Hauth, A., Millican, K., et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Team, G., Riviere, M., Pathak, S., Sessa, P. G., Hardin, C., Bhupatiraju, S., Hussenot, L., Mesnard, T., Shahriari, B., Ramé, A., et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- Tong, P., Brown, E., Wu, P., Woo, S., IYER, A. J. V., Akula, S. C., Yang, S., Yang, J., Middepogu, M., Wang, Z., et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. In *Advances in Neural Information Processing Systems*, volume 37, pp. 87310–87356, 2024a.
- Tong, Y., Zhang, X., Wang, R., Wu, R., and He, J. DART-math: Difficulty-aware rejection tuning for mathematical problem-solving. In *Advances in Neural Information Processing Systems*, 2024b.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023b.
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 353–355, November 2018.
- Wang, K., Dimitriadis, N., Ortiz-Jimenez, G., Fleuret, F., and Frossard, P. Localizing task information for improved model merging and compression. In *International Conference on Machine Learning*, 2024a.
- Wang, Y., Shi, H., Han, L., Metaxas, D., and Wang, H. Blob: Bayesian low-rank adaptation by backpropagation for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pp. 67758–67794, 2024b.
- Wang, Z., Liang, J., He, R., Wang, Z., and Tan, T. LoRA-pro: Are low-rank adapters properly optimized? In *International Conference on Learning Representations*, 2025.
- Wei, Y., Wang, Z., Liu, J., Ding, Y., and ZHANG, L. Magi-coder: Empowering code generation with OSS-instruct. In *International Conference on Machine Learning*, 2024.
- Weyssow, M., Zhou, X., Kim, K., Lo, D., and Sahraoui, H. Exploring parameter-efficient fine-tuning techniques for code generation with large language models. *ACM Transactions on Software Engineering and Methodology*, 34(7):1–25, 2025.

- Williams, A., Nangia, N., and Bowman, S. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of the 2018 conference of the North American chapter of the association for computational linguistics: human language technologies*, pp. 1112–1122, 2018.
- Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Lopes, R. G., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., and Schmidt, L. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, volume 162, pp. 23965–23998, 2022.
- Xiong, S., Zhao, Y., Zhang, J., Mengxiang, L., He, Z., Li, X., and Song, S. Dual prompt tuning based contrastive learning for hierarchical text classification. In *Findings of the association for computational linguistics ACL 2024*, pp. 12146–12158, 2024.
- Yadav, P., Tam, D., Choshen, L., Raffel, C. A., and Bansal, M. Ties-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, 2023.
- Yan, A., McAuley, J., Lu, X., Du, J., Chang, E. Y., Gentili, A., and Hsu, C.-N. Radbert: Adapting transformer-based language models to radiology. *Radiology: Artificial Intelligence*, 4(4):e210258, 2022.
- Yang, E., Wang, Z., Shen, L., Liu, S., Guo, G., Wang, X., and Tao, D. Adamerging: Adaptive model merging for multi-task learning. In *International Conference on Learning Representations*, 2024.
- Yu, L., Yu, B., Yu, H., Huang, F., and Li, Y. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *International Conference on Machine Learning*, 2024.
- Yu, X., Zhou, C., Kuai, Z., Zhang, X., and Fang, Y. Gcot: Chain-of-thought prompt learning for graphs. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pp. 3669–3679, 2025.
- Yun, P. and Liu, M. Laplace approximation based epistemic uncertainty estimation in 3d object detection. In *6th Annual Conference on Robot Learning*, 2022.
- Zhang, F. Z., Albert, P., Rodriguez-Opazo, C., van den Hengel, A., and Abbasnejad, E. Knowledge composition using task vectors with learned anisotropic scaling. In *Advances in Neural Information Processing Systems*, volume 37, pp. 67319–67354, 2024.
- Zhang, J., Liu, J., He, J., et al. Composing parameter-efficient modules with arithmetic operation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 12589–12610, 2023a.
- Zhang, S., Dong, L., Li, X., Zhang, S., Sun, X., Wang, S., Li, J., Hu, R., Zhang, T., Wang, G., et al. Instruction tuning for large language models: A survey. *ACM Computing Surveys*, 2023b.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623, 2023.
- Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., and Hou, L. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.
- Zhu, B., Niu, Y., Han, Y., Wu, Y., and Zhang, H. Prompt-aligned gradient for prompt tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15659–15669, 2023.
- Zhu, Z., Hong, J., and Zhou, J. Data-free knowledge distillation for heterogeneous federated learning. In *International Conference on Machine Learning*, pp. 12878–12889, 2021.

A. Related Work

Transfer Learning in NLP. Fine-tuning LLMs pre-trained on massive corpora has become the dominant transfer learning paradigm in NLP, enabling a single pre-trained backbone to effectively adapt to many downstream tasks (Howard & Ruder, 2018; Yu et al., 2024; Zhang et al., 2023b). However, full-parameter fine-tuning is computationally expensive and memory-intensive, especially as model sizes continue to grow. Parameter-efficient fine-tuning has emerged as a promising solution to this problem (Fu et al., 2023; Ding et al., 2023; Zhang et al., 2023a). Existing approaches can be broadly categorized into three classes. First, prompt-based tuning learns additional prompts to steer a frozen backbone, thereby changing model behavior with minimal extra parameters (Lester et al., 2021; Shi & Lipani, 2023; Zhu et al., 2023; Xiong et al., 2024; Yu et al., 2025). Second, adapter-based tuning inserts small, task-specific modules into each layer of the network and trains only these modules while keeping the original weights fixed (Houlsby et al., 2019; He et al., 2021; Sung et al., 2022; Guo et al., 2024; Cai et al., 2025). Third, Low-Rank Adaptation (LoRA) approximates full-rank weight updates using the product of two low-rank matrices, significantly reducing the number of trainable parameters while preserving expressiveness (Hu et al., 2022; Liu et al., 2024b; Meng et al., 2024; Hayou et al., 2024; Borse et al., 2024; Wang et al., 2024b; 2025). In this paper, we focus on a complementary question: how to transfer fine-tuning knowledge from an old model to an updated one. Simply fine-tuning the new model on the original task data can be prohibitively costly. Moreover, when the original fine-tuning data are no longer accessible due to privacy, compliance, or storage constraints, such re-fine-tuning is not even feasible. This motivates the need for a transfer mechanism that does not require re-running fine-tuning on the updated model, but instead leverages the already fine-tuned old model to propagate task knowledge to newer checkpoints.

Model Merging. Model merging aims to integrate multiple fine-tuned models into a single model directly in parameter space, while retaining each original model’s capabilities (Matena & Raffel, 2022; Jin et al., 2023; Yang et al., 2024). Model soups (Wortsman et al., 2022) obtain a more robust model by averaging checkpoints that share the same pre-trained initialization; their success is attributed to an empirical phenomenon that fine-tuned models from a shared initialization tend to lie in the same low-error basin of the loss landscape (Neyshabur et al., 2020). Task arithmetic (Ilharco et al., 2023) further treats the parameter difference (delta) between fine-tuned models and the base model as a task vector, modulating model behavior through arithmetic combinations like addition and subtraction. Subsequent work applies conflict-aware rules to deltas before aggregation, such as clipping smaller changes, resolving sign conflicts through consistency selection (Yadav et al., 2023; Huang et al., 2024), or mitigating parameter update interference across tasks via sparse corrections (Yu et al., 2024; Wang et al., 2024a; Gargiulo et al., 2025; Iurada et al., 2025). Additionally, alignment-based methods attempt to synchronize neural or channel representations between independently trained models to address permutation symmetry and representational inconsistencies, enabling fusion across independently trained models (Ainsworth et al., 2023; Guerrero-Peña et al., 2023; Stoica et al., 2024; Crisostomi et al., 2024; Imfeld et al., 2024). These works aim to merge multiple fine-tuned checkpoints—trained for different downstream tasks—into a single model that preserves the combined set of capabilities. In contrast, fine-tuning transfer reuses fine-tuning knowledge to adapt a different base model, with the goal of producing the task-adapted version of that new base model.

Recent work aims to transfer knowledge gained through fine-tuning to updated pre-trained models (Rinaldi et al., 2025a,b; Cao et al., 2025). These methods primarily operate in parameter space: they compute parameter correction vectors by comparing fine-tuned models with their base checkpoints, then apply these corrections to the new model. However, different pre-trained models usually yield distinct loss landscapes, and even re-basin or rescaling strategies struggle to adjust for such differences in parameter space. As a result, the same downstream task may not correspond to consistent update behaviors across different models. In contrast, we adopt a Bayesian-updating perspective on fine-tuning and explicitly separate task effects from the choice of base model. We introduce a reusable task-update factor that is invariant across base models and has a fixed-dimensional parameterization, characterize when such factors exist, and propose a method that enables transferring fine-tuning knowledge to new models without post-training.

B. Limitations

Although BTransfer provides a principled alternative to parameter-space transfer, it has several limitations.

Computation. Our instantiation for deep networks relies on a Laplace approximation, which requires estimating diagonal Fisher information matrices. While our sensitivity study indicates that a few hundred samples are sufficient to reach near-optimal performance, Fisher estimation still incurs non-trivial overhead compared to purely parameter-space methods such as simple parameter differencing. This overhead can become substantial for very large models or when multiple task updates must be handled jointly. Exploring more efficient approximations or alternatives to the Fisher matrix is a natural

direction to mitigate this cost.

Local approximation. Our deep-network instantiation of `BTransfer` relies on a Laplace lift, which approximates the posterior locally around a converged solution with a Gaussian distribution. Accordingly, this instantiation should be viewed as a local approximation rather than a guarantee of global transferability. Its transfer quality depends on how accurately the Laplace approximation captures the posterior geometry in the neighborhood of that solution. Extending the current local lift to one with a broader region of validity is an important and promising direction for future work.

Modeling. Our current method assumes that the old and new base models share the same architecture, so that their priors and posteriors can be embedded into a common exponential-family representation. This “shared exponential family” setting holds naturally when transferring across different checkpoints of a fixed backbone, but does not directly extend to transfers between heterogeneous architectures—a limitation shared by many parameter-space methods. Designing update factors that remain reusable even when the underlying architecture changes substantially is an important direction for future work.

C. Proofs of Theorem and Corollary

C.1. Proof of Theorem 3.1

Theorem C.1 (Theorem 3.1 (restated)). *Let Δ denote the set of all probability distributions over θ for which $\mathbb{E}_q[|g|] < \infty$. Assume there exist (ϕ, λ) and a constant c such that $\mathbb{E}_{\hat{q}_i}[g(\theta)] = c$ for all $p_i \in \mathcal{P}$. If we define the variational family in Eq. (1) as $\mathcal{Q}_c := \{q \in \Delta : \mathbb{E}_q[g(\theta)] \geq c\}$, then:*

1. **Existence.** *There exists a reusable task-update factor, namely $\mathcal{U}(\theta; D_C) = \exp(\lambda^\top \phi(\theta))$.*
2. **Maximality.** *For any strictly larger convex set $\mathcal{Q}' \supsetneq \mathcal{Q}_c$, this reusable task-update factor no longer exists.*

Proof. Let us briefly recall the notation introduced in the main text. For each prior $p_i \in \mathcal{P}$, the variational fine-tuning objective in Eq. (1) is

$$F_i(q) := \mathbb{E}_{q(\theta)}[-\log p(D_C | \theta)] + \text{KL}(q(\theta) \| p_i(\theta)). \quad (6)$$

Fix basis functions $\phi(\theta) = (\phi_1(\theta), \dots, \phi_d(\theta))$ and a parameter vector $\lambda = \lambda(D_C) \in \mathbb{R}^d$. Define

$$\hat{q}_i(\theta) := \frac{\exp(\phi(\theta)^\top \lambda) p_i(\theta)}{Z_i}, \quad Z_i := \int \exp(\phi(\theta)^\top \lambda) p_i(\theta) d\theta,$$

and the auxiliary function

$$g(\theta) := \phi(\theta)^\top \lambda - \log p(D_C | \theta).$$

By assumption, there exists a constant c such that

$$\mathbb{E}_{\hat{q}_i}[g(\theta)] = c, \quad \forall p_i \in \mathcal{P}. \quad (7)$$

We further define the halfspace variational family

$$\mathcal{Q}_c := \{q \in \Delta : \mathbb{E}_q[g(\theta)] \geq c\}. \quad (8)$$

Existence. Expanding Eq. (6) yields

$$F_i(q) = \int q(\theta) \left(-\log p(D_C | \theta) + \log q(\theta) - \log p_i(\theta) \right) d\theta.$$

For $q(\theta) > 0$ a.e., the variational derivative is

$$\frac{\delta F_i}{\delta q}(\theta) = -\log p(D_C | \theta) + \log q(\theta) - \log p_i(\theta) + 1.$$

Using $\log \hat{q}_i(\theta) = \phi(\theta)^\top \lambda + \log p_i(\theta) - \log Z_i$, we obtain

$$\left. \frac{\delta F_i}{\delta q}(\theta) \right|_{q=\hat{q}_i} = g(\theta) + (1 - \log Z_i), \quad (9)$$

namely, the derivative at $\hat{q}_i$ equals $g(\theta)$ plus a θ -independent constant.

Since $F_i(\cdot)$ is convex in q , the first-order convexity inequality implies that for any $q \in \Delta$,

$$F_i(q) \geq F_i(\hat{q}_i) + \int \frac{\delta F_i}{\delta q}(\theta) \Big|_{q=\hat{q}_i} (q(\theta) - \hat{q}_i(\theta)) d\theta. \quad (10)$$

Substituting Eq. (9) into Eq. (10), we obtain

$$F_i(q) - F_i(\hat{q}_i) \geq \int g(\theta)(q(\theta) - \hat{q}_i(\theta)) d\theta = \mathbb{E}_q[g(\theta)] - \mathbb{E}_{\hat{q}_i}[g(\theta)], \quad (11)$$

where the constant term is eliminated using $\int (q - \hat{q}_i) d\theta = 0$.

By Eq. (7) and Eq. (8), we have $\mathbb{E}_q[g(\theta)] \geq c = \mathbb{E}_{\hat{q}_i}[g(\theta)]$. Plugging this into Eq. (11) gives $F_i(q) \geq F_i(\hat{q}_i)$ for all $q \in \mathcal{Q}_c$, and hence

$$\hat{q}_i \in \arg \min_{q \in \mathcal{Q}_c} F_i(q), \quad \forall p_i \in \mathcal{P}.$$

Moreover, each minimizer satisfies $\hat{q}_i(\theta) \propto \mathcal{U}(\theta; D_C) p_i(\theta)$ with the same $\mathcal{U}(\theta; D_C) = \exp(\phi(\theta)^\top \lambda)$ for all $p_i \in \mathcal{P}$. This proves the existence of a reusable task-update factor satisfying Defs. 3.1 and 3.2.

Maximality. It remains to establish maximality relative to the fixed pair (ϕ, λ) . Let $\mathcal{Q}' \supsetneq \mathcal{Q}_c$ be any strictly larger convex set. Then, there exists $q' \in \mathcal{Q}' \setminus \mathcal{Q}_c$ such that

$$\mathbb{E}_{q'}[g(\theta)] < c. \quad (12)$$

Since $\hat{q}_i \in \mathcal{Q}_c \subset \mathcal{Q}'$, convexity of $\mathcal{Q}'$ implies that the mixture

$$q_t := (1-t)\hat{q}_i + tq', \quad t \in (0, 1),$$

also belongs to $\mathcal{Q}'$.

Suppose that $\hat{q}_i$ were a minimizer of $F_i(\cdot)$ over $\mathcal{Q}'$. Because $\mathcal{Q}'$ is convex, a first-order necessary condition for optimality is that the directional derivative along any feasible direction is nonnegative. In particular, along $q' - \hat{q}_i$, we must have

$$\frac{d}{dt} F_i(q_t) \Big|_{t=0^+} \geq 0. \quad (13)$$

On the other hand, using Eq. (9) and $\int (q' - \hat{q}_i) = 0$, we compute

$$\begin{aligned} \frac{d}{dt} F_i(q_t) \Big|_{t=0^+} &= \int \frac{\delta F_i}{\delta q}(\theta) \Big|_{q=\hat{q}_i} (q'(\theta) - \hat{q}_i(\theta)) d\theta \\ &= \int g(\theta)(q'(\theta) - \hat{q}_i(\theta)) d\theta \\ &= \mathbb{E}_{q'}[g(\theta)] - \mathbb{E}_{\hat{q}_i}[g(\theta)] \\ &= \mathbb{E}_{q'}[g(\theta)] - c < 0, \end{aligned} \quad (14)$$

where the last inequality follows from Eq. (12) and Eq. (7). The inequality Eq. (14) contradicts the necessary condition Eq. (13), and thus $\hat{q}_i$ cannot be a minimizer of $F_i(\cdot)$ over $\mathcal{Q}'$.

Equivalently, the update factor $\mathcal{U}(\theta; D_C) = \exp(\phi(\theta)^\top \lambda)$ fails to remain reusable once the feasible set is enlarged to any strictly larger convex $\mathcal{Q}'$. This proves the maximality statement relative to the fixed pair (ϕ, λ) . $\square$

C.2. Proof of Corollary 3.1

Corollary C.1 (Corollary 3.1 (restated)). *Suppose there exist basis functions $\phi(\theta)$, parameter λ , and a constant $\kappa(D_C)$ such that, for all θ , $\lambda^\top \phi(\theta) + \kappa(D_C) \geq \log p(D_C | \theta)$. Under the conditions of Thm. 3.1, the maximal variational family induced by the surrogate is*

$$\mathcal{Q}_c = \{q \in \Delta : \mathbb{E}_q[r(\theta)] \geq \bar{r}\},$$

where the residual is $r(\theta) := \lambda^\top \phi(\theta) + \kappa(D_C) - \log p(D_C | \theta)$ and $\bar{r} := \mathbb{E}_{\hat{q}_i}[r(\theta)]$ is a constant independent of i .

Proof. Recall that $g(\theta) = \lambda^\top \phi(\theta) - \log p(D_C | \theta)$. By definition of $r(\theta)$,

$$r(\theta) = g(\theta) + \kappa(D_C). \quad (15)$$

Under the conditions of Thm. 3.1, there exists a constant c such that $\mathbb{E}_{\hat{q}_i}[g(\theta)] = c$ for all $p_i \in \mathcal{P}$, where $\hat{q}_i(\theta) \propto \exp(\lambda^\top \phi(\theta))p_i(\theta)$. By Eq. (15) and $\mathbb{E}_{\hat{q}_i}[g] = c$, there exists

$$\bar{r} := \mathbb{E}_{\hat{q}_i}[r(\theta)] = \mathbb{E}_{\hat{q}_i}[g(\theta)] + \kappa(D_C) = c + \kappa(D_C), \quad (16)$$

which is independent of i .

Finally, the maximal variational family in Thm. 3.1 is

$$\mathcal{Q}_c = \{q \in \Delta : \mathbb{E}_q[g(\theta)] \geq c\}.$$

Using Eq. (15), the inequality $\mathbb{E}_q[g(\theta)] \geq c$ is equivalent to

$$\mathbb{E}_q[r(\theta)] - \kappa(D_C) \geq c \iff \mathbb{E}_q[r(\theta)] \geq c + \kappa(D_C).$$

By Eq. (16), we have

$$\mathcal{Q}_c = \{q \in \Delta : \mathbb{E}_q[r(\theta)] \geq \bar{r}\},$$

which is the claimed form. $\square$

C.3. Proof of Corollary 3.2

Corollary C.2 (Corollary 3.2 (restated)). *Assume all priors $p_i \in \mathcal{P}$ belong to a common exponential family and, for the dataset D_C , the Bayesian posterior under each p_i also remains within that family. Then the maximal variational family satisfies $\mathcal{Q}_c = \Delta$.*

Proof. Let $\mathcal{E}$ be a common exponential family over θ with a *minimal* natural-parameterization $(h(\theta), T(\theta), A(\cdot))$. In particular, every distribution in $\mathcal{E}$ admits a unique natural-parameter representation

$$p(\theta; \eta) = h(\theta) \exp(\eta^\top T(\theta) - A(\eta)).$$

Since each prior $p_i \in \mathcal{P}$ belongs to $\mathcal{E}$, there exists a natural parameter α_i such that

$$p_i(\theta) = h(\theta) \exp(\alpha_i^\top T(\theta) - A(\alpha_i)). \quad (17)$$

Moreover, for the dataset D_C , the corresponding Bayesian posterior also remains in $\mathcal{E}$. Hence, there exists β_i such that

$$p_i(\theta | D_C) = h(\theta) \exp(\beta_i^\top T(\theta) - A(\beta_i)). \quad (18)$$

Since $p_i(\theta | D_C) \propto p(D_C | \theta) p_i(\theta)$, combining Eq. (17)–Eq. (18) yields

$$\log p(D_C | \theta) = (\beta_i - \alpha_i)^\top T(\theta) + \kappa_i(D_C), \quad \forall \theta, \quad (19)$$

where $\kappa_i(D_C)$ is a constant independent of θ .

Notice that the left-hand side of Eq. (19) does not depend on i . Thus, for any i, j , we have

$$(\beta_i - \alpha_i)^\top T(\theta) + \kappa_i(D_C) = (\beta_j - \alpha_j)^\top T(\theta) + \kappa_j(D_C), \quad \forall \theta.$$

Rearranging gives

$$((\beta_i - \alpha_i) - (\beta_j - \alpha_j))^\top T(\theta) = \kappa_j(D_C) - \kappa_i(D_C), \quad \forall \theta. \quad (20)$$

The right-hand side of Eq. (20) is θ -independent. By minimality of the representation, if $v^\top T(\theta)$ is constant a.e. in θ , then $v = 0$. Therefore, we have

$$(\beta_i - \alpha_i) - (\beta_j - \alpha_j) = 0 \quad \text{and} \quad \kappa_i(D_C) = \kappa_j(D_C), \quad \forall i, j.$$

That is, there exist η and $\kappa(D_C)$, both independent of i , such that

$$\log p(D_C | \theta) = \eta^\top T(\theta) + \kappa(D_C), \quad \forall \theta.$$

Taking $\lambda = \eta$ and $\phi(\theta) = T(\theta)$, we have $g(\theta) = \lambda^\top \phi(\theta) - \log p(D_C | \theta) = -\kappa(D_C)$, which is a θ -independent constant. Hence, we have $\mathbb{E}_q[g(\theta)] = -\kappa(D_C)$ for every $q \in \Delta$, and the defining inequality $\mathbb{E}_q[g(\theta)] \geq c$ holds for all $q \in \Delta$ with $c = -\kappa(D_C)$. This establishes that $\mathcal{Q}_c = \Delta$, which completes the proof. $\square$

D. Derivation of the Lift Step

Here, we provide the detailed derivation for lifting a point estimate θ_m to a distribution $\pi_m(\theta)$ using the Laplace approximation (Daxberger et al., 2021), and subsequently reformulating it into the exponential-family form presented in Eq. (4).

Let $\mathcal{L}(\theta)$ denote training loss, and θ_m be a local minimum of $\mathcal{L}(\theta)$, the Laplace approximation constructs a Gaussian approximation to posterior by performing a second-order Taylor expansion of $\mathcal{L}(\theta)$ around θ_m :

$$\mathcal{L}(\theta) \approx \mathcal{L}(\theta_m) + \nabla \mathcal{L}(\theta_m)^\top (\theta - \theta_m) + \frac{1}{2} (\theta - \theta_m)^\top \nabla^2 \mathcal{L}(\theta_m) (\theta - \theta_m).$$

Since θ_m is the local minimum, the gradient vanishes, i.e., $\nabla \mathcal{L}(\theta_m) = \mathbf{0}$. Letting $\Lambda_m \triangleq \nabla^2 \mathcal{L}(\theta_m)$ denote the precision matrix, the approximate posterior density is given by:

$$\pi_m(\theta) \propto \exp \left(-\frac{1}{2} (\theta - \theta_m)^\top \Lambda_m (\theta - \theta_m) \right).$$

Expanding the quadratic term gives

$$-\frac{1}{2} (\theta - \theta_m)^\top \Lambda_m (\theta - \theta_m) = -\frac{1}{2} (\theta^\top \Lambda_m \theta - \theta^\top \Lambda_m \theta_m - \theta_m^\top \Lambda_m \theta + \theta_m^\top \Lambda_m \theta_m).$$

Using the symmetry of the Hessian matrix Λ_m , which implies $\theta^\top \Lambda_m \theta_m = \theta_m^\top \Lambda_m \theta$, and absorbing the constant term $-\frac{1}{2} \theta_m^\top \Lambda_m \theta_m$ into the normalization constant, we obtain

$$\log \pi_m(\theta) = -\frac{1}{2} \theta^\top \Lambda_m \theta + (\Lambda_m \theta_m)^\top \theta + \text{const.}$$

Next, we use the trace operator and the vectorization identity $\text{tr}(A^\top B) = \text{vec}(A)^\top \text{vec}(B)$ to linearize the quadratic term w.r.t. the sufficient statistics:

$$\begin{aligned} -\frac{1}{2} \theta^\top \Lambda_m \theta &= \text{tr} \left(-\frac{1}{2} \Lambda_m \theta \theta^\top \right) \\ &= \text{vec} \left(-\frac{1}{2} \Lambda_m \right)^\top \text{vec}(\theta \theta^\top). \end{aligned}$$

Substituting this back into the log-density yields:

$$\log \pi_m(\theta) = (\Lambda_m \theta_m)^\top \theta + \text{vec} \left(-\frac{1}{2} \Lambda_m \right)^\top \text{vec}(\theta \theta^\top) + \text{const.}$$

Concatenating the parameter and statistic vectors gives

$$\pi_m(\theta) \propto \exp \left(\begin{bmatrix} \Lambda_m \theta_m \\ \text{vec}(-\frac{1}{2} \Lambda_m) \end{bmatrix}^\top \begin{bmatrix} \theta \\ \text{vec}(\theta \theta^\top) \end{bmatrix} \right).$$

Defining $b_m = \Lambda_m \theta_m$, this recovers Eq. (4).

E. Experiment Details and Additional Experiments

E.1. Computational cost

To evaluate the computational efficiency of BTransfer, we compare its runtime against full fine-tuning (to obtain θ_{new}^C) and parameter-space transfer baselines on the legal and mental backbones. Reported runtimes exclude data preparation and evaluation, and measure only method-specific computation. For BTransfer, we vary the Fisher sample size over $\{500, 2,500, 5,000\}$. As shown in Tabs. 6 and 7, parameter-space methods are inexpensive, requiring about 0.3s for Param Δ and 8–12s for TransFusion. BTransfer requires 14–117s for curvature estimation, yet remains substantially cheaper than full fine-tuning at roughly 2,000–3,200s. Although more expensive than parameter-space methods, BTransfer offers a favorable trade-off between computational cost and transfer quality.

Table 6. Runtime comparison on `legal`. We report the computational cost in seconds for full fine-tuning (FT), parameter-space baselines, and BTransfer.

Dataset	FT	Param Δ	TransFusion	BTransfer (s)		
	(s)	(s)	(s)	500	2500	5000
SST-2	1995.84	0.37	11.21	14.01	57.68	114.11
MNLI	3165.12	0.36	12.24	15.82	61.13	117.55
QNLI	1990.08	0.33	11.10	14.81	59.57	116.23
RTE	2072.88	0.38	11.11	14.02	56.64	90.24
Subj	2117.88	0.33	11.26	13.89	59.86	116.03
MPQA	2075.04	0.39	10.86	14.00	59.33	115.62
MR	2043.00	0.32	11.39	13.88	59.33	116.06

Table 7. Runtime comparison on `mental`. We report the computational cost in seconds for full fine-tuning (FT), parameter-space baselines, and BTransfer.

Dataset	FT	Param Δ	TransFusion	BTransfer (s)		
	(s)	(s)	(s)	500	2500	5000
SST-2	1995.84	0.32	8.82	14.17	58.98	116.10
MNLI	3103.92	0.31	9.21	15.95	60.28	116.59
QNLI	2008.08	0.32	8.31	14.57	59.46	116.68
RTE	2124.00	0.34	8.98	13.84	56.96	90.28
Subj	2256.12	0.37	9.26	14.06	59.58	117.02
MPQA	2025.00	0.37	9.12	13.92	59.23	115.35
MR	2052.00	0.33	8.25	13.82	59.40	115.61

E.2. Evaluation of source-domain performance

In transfer settings, retaining source-domain performance is also important. To evaluate source-domain retention, we take `legal-roberta-base` as an example and use three LegalLAMA-style probing tasks (Chalkidis et al., 2023): ECHR Articles (EA), Contractual Section Titles (CS), and Contract Types (CT). These tasks assess legal-domain knowledge, and we report ranking-based metrics, including Mean Reciprocal Rank (MRR) and Hits@5.

As shown in Tab. 8, all transfer methods degrade relative to the original legal model θ_{new} . However, BTransfer best preserves source-domain performance on average, achieving the highest average MRR and Hits@5, with 21.60 and 32.48, compared with 21.42/31.94 for Param Δ and 21.42/31.91 for TransFusion. These results indicate that BTransfer better retains source-domain knowledge than parameter-space transfer baselines.

E.3. Normalized Gain Metric

For joint transfer over a task set T , let $a_{T,t}^{(m)}$ denote the accuracy of method $m \in \{\theta_{new}, \theta_{old}^C, \text{Param}\Delta, \text{BTransfer}\}$ on task $t \in T$. We first normalize performance on each task by linearly mapping the new pre-trained model θ_{new} to 0 and the old fine-tuned model θ_{old}^C to 1:

$$r_{T,t}^{(m)} = \frac{a_{T,t}^{(m)} - a_{T,t}^{(\theta_{new})}}{a_{T,t}^{(\theta_{old}^C)} - a_{T,t}^{(\theta_{new})}},$$

so that $r_{T,t}^{(\theta_{new})} = 0$ and $r_{T,t}^{(\theta_{old}^C)} = 1$ for all t .

We then average over tasks in T to obtain the normalized gain under joint transfer:

$$R_T^{(m)} = \frac{1}{|T|} \sum_{t \in T} r_{T,t}^{(m)}.$$

Thus, $R_T^{(m)} = 0$ means matching θ_{new} on average, and $R_T^{(m)} = 1$ means matching θ_{old}^C .

Table 8. Results on three LegalLAMA-style probing tasks. We report MRR (%) and Hits@5 (%) for the original legal model θ_{new} , parameter-space baselines (Param Δ , TransFusion), and BTransfer (ours). For θ_{new} , each probing task has a single source-domain score because the model is not fine-tuned on the target tasks; for transfer methods, each row evaluates the transferred model obtained for the corresponding target task.

Method		θ_{new}		Param Δ		TransFusion		BTransfer	
Metric		MRR	Hits@5	MRR	Hits@5	MRR	Hits@5	MRR	Hits@5
EA	SST-2	39.45	50.12	24.70	38.06	24.69	38.18	24.45	37.32
	MNLI	-	-	24.82	39.99	24.82	39.84	24.75	40.53
	QNLI	-	-	24.26	38.01	24.28	38.06	24.45	36.99
	RTE	-	-	24.51	38.25	24.54	38.25	25.09	39.11
	Subj	-	-	25.33	38.98	25.35	38.97	25.44	38.28
	MPQA	-	-	25.30	38.51	25.27	38.49	24.39	38.78
	MR	-	-	24.19	38.27	24.18	38.21	24.68	38.26
CS	SST-2	18.10	25.38	18.04	24.19	18.02	24.18	18.78	25.65
	MNLI	-	-	17.99	25.69	18.01	25.64	17.57	24.38
	QNLI	-	-	17.08	22.74	17.10	22.79	16.70	23.44
	RTE	-	-	17.44	23.43	17.50	23.49	17.98	24.89
	Subj	-	-	17.63	23.65	17.62	23.65	17.72	22.53
	MPQA	-	-	18.31	23.00	18.35	22.90	17.96	25.00
	MR	-	-	18.21	23.20	18.24	23.41	17.89	25.00
CT	SST-2	25.56	40.78	21.77	29.74	21.65	29.72	22.29	34.99
	MNLI	-	-	21.29	32.29	21.28	32.27	22.34	34.97
	QNLI	-	-	21.20	36.26	21.17	35.80	21.72	36.98
	RTE	-	-	21.52	34.87	21.61	34.58	22.35	32.87
	Subj	-	-	21.11	34.17	21.12	34.19	22.80	36.31
	MPQA	-	-	22.62	32.97	22.69	33.00	22.26	33.33
	MR	-	-	22.42	34.53	22.32	34.51	21.93	32.47
AVG		27.70	38.76	21.42	31.94	21.42	31.91	21.60	32.48

E.4. Large Language Model Datasets.

The dataset and metric configurations used for the large language model transfer experiments are as follows. For Fisher estimation at the old and new pre-trained checkpoints (θ_{old} and θ_{new}), we sample from the English Wikipedia 20220301 dump. For each capability, the training set for Fisher estimation listed below is used to estimate the Fisher at the corresponding fine-tuned checkpoint θ_{old}^C .

- **Coding.** Training Set for Fisher Estimation: Magicoder (Wei et al., 2024). Evaluation sets: HumanEval+ and MBPP+ (Liu et al., 2023). Metric: Pass@1. We report the mean Pass@1 over the two evaluation sets.
- **Safety.** Training Set for Fisher Estimation: WildGuardMix (Han et al., 2024). Evaluation sets: WildGuardTest (Han et al., 2024), HarmBench (Mazeika et al., 2024), and XSTest (Röttger et al., 2024). Metrics: RTA on WildGuardTest and HarmBench, and Accuracy on XSTest. We report the mean over these three metrics.
- **Math.** Training Set for Fisher Estimation: DART-Math (Tong et al., 2024b). Evaluation set: GSM8k (Cobbe et al., 2021). Metric: EM under 8-shot CoT.
- **Multilingual.** Training Set for Fisher Estimation: Aya (Singh et al., 2024). Evaluation set: M-ARC (Lai et al., 2023). Metric: Normalized Accuracy.
- **Instruction-following.** Training Set for Fisher Estimation: TULU-3 persona IF (Lambert et al., 2025). Evaluation set: IFEval (Zhou et al., 2023). Metric: prompt-level accuracy.